

Puneet Ludu

ML Engineer & Tech Lead

Production ML · Applied AI + Agentic · 13+ years

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 [Google Scholar](#)

 No sponsorship required (LPR)

Experience

13+ years

Zillow (Zestimate)

Machine Learning Engineer (Tech Lead) · Remote

5Y

Sep 2021 – Present

Next-Gen Zestimate: Explainable Valuation (PyTorch, Databricks) [Glass-box Modeling]

- Led architecture experiments and production evaluation for an explainable valuation model with per-feature dollar contributions; improved accuracy through learned comp weighting and evaluated explanation stability nationally.
- Impact:** Building Zillow's first explainable Zestimate, with per-feature dollar attribution for downstream valuation products.

Property Condition & Quality Image Embeddings (PyTorch, Databricks, CLIP)

- Led research and model refinement for CLIP-based property scoring on 1.95M homes: **~70% five-class accuracy and ~0.91 weighted AUC** on held-out validation. Scored 16.5M home-listing combinations in staging.
- Impact:** Scoring pipeline in production; Zestimate integration pending.

Zestimate Customer Care Agent (LangChain, FastAPI) [RAG + Custom Judges + Bounded Output]

- Built one of Zillow's first agentic AI tools for customer care, with RAG and real-time services exposing property history, features, and comps across ~150M homes.
- Encoded expert triage as agent tools; reduced hallucinations with custom judges, curated context, and bounded responses.
- Answered ~50–60% of questions in internal evaluation. Piloted with care reps reviewing explanations before customer use.

Listing IQ: Interactive CMA Platform (Django, DocumentDB, H3 Geospatial) [Real-time Comps + Valuation]

- Led two engineers from interactive CMA prototype through handoff to ShowingTime, owning the ML architecture and incorporating seller-agent feedback; productionized the real-time ML services.
 - Moved comp selection from batch to real time using geospatial retrieval and ranking. Property edits update comps and valuations immediately; full reports in under 15 seconds.
- Impact:** Shipped as Listing IQ CMA, bringing Zestimate valuation data into a commercial seller-agent product.

Active Listing Comps Engine (Databricks, PySpark, H3 Geospatial) [Similarity Ranking + Batch Pipeline]

- Architected the daily similarity-ranking pipeline for 3M+ active listings behind Showcase's Listing Performance Dashboard.
 - Built deduplication and lifecycle tracking across active, sold, and removed listings to maintain an accurate comp universe.
 - Led a data source migration that expanded listing coverage from **90% to 98.5%**.
- Impact:** **95% reduction in listing-visibility complaints** following the coverage improvement.

Infrastructure & Engineering Leadership (FastAPI, AWS, Terraform, Databricks, PySpark) [API Gateway + Legacy Migration]

- Built a FastAPI reverse-proxy gateway decoupling downstream applications from model changes; designed for 20–40M req/day, with A/B testing support and a 99.5% SLA.
 - Led legacy valuation retirement and Redis-to-feature-store migration; discovered two undocumented downstream consumers, rolled back, and revised the rollout.
 - Led training ETL migration from Metaflow/Kubernetes to Databricks PySpark. Platform-specific optimization reversed an initial 8× latency and 10× cost regression, restoring production parity.
- Impact:** **\$350K annual savings** from decommissioning the legacy valuation stack, including Redis.

Mentorship & Technical Leadership

Managed a summer intern (2023) and mentored 4+ engineers; authored RFCs and architecture docs adopted as reference patterns outside the team; conducted technical interviews.

Match Group (OkCupid)

Machine Learning Engineer · New York City

1.5Y

May 2020 – Sep 2021

Discount Optimization (Python, Keras, TensorFlow, Weights & Biases) [Wide&Deep + Uncertainty Estimation]

- Owned the subscription discount optimization pipeline: feature engineering under PII constraints, model training, A/B testing, deployment, and production monitoring.
 - Diagnosed epistemic uncertainty across random seeds and built a 100-model ensemble to stabilize discount recommendations across retraining runs.
- Impact:** **6% revenue increase** vs. baseline in A/B testing.

Experience

Continued

FactSet

Machine Learning Engineer · New York City

5Y

Apr 2015 – May 2020

ML-Powered Financial Data Extraction (Python, TensorFlow, Keras, SageMaker) [CNN, ELMo, BiLSTM]

- Led development of a speaker identification system for earnings calls using spectrogram-based CNNs.
- Built private company fact extraction pipeline across 1.6M websites using ELMo/BiLSTM; rewrote the language identification service.
- Led machine translation infrastructure for financial content (Polish SMT achieving BLEU 69.10).

Impact: 20% reduction in annotation time in a limited production rollout of earnings-call speaker identification 

Financial Document Search & Ranking Systems (Apache Spark, Java, Python) [Distributed Trie, N-gram LM, Vector Space]

- Led team of 3 engineers on Document Screening: built autosuggestion and concept similarity systems.
- Built the FingerPrinter deduplication service, reducing response time from 1000ms to 100ms (10×).
- Architected Formula Lookup using distributed trie and n-gram language models on Spark.

Impact: Formula ranking improved from 5.6 to 2.3; document processing was 66% faster. Systems supported StreetAccount trending news.

Tata Research Development and Design Centre

ML Research Engineer · India

2Y

Jul 2011 – Jul 2013

Event Detection in Time Series (Java, Python, RapidMiner) [SVM-RBF]

- Wrote a Shape Context-based algorithm for detecting recurring patterns in time series; matched SAX/DTW accuracy on general benchmarks and outperformed by 7% in the car sensor domain.

Data Harmonization Framework (DHF) (Java, Apache Pig)

- Built a MapReduce ETL framework that combines enterprise data from multiple sources in near real time. Published at IEEE INDIN 2013.

Skills

Languages	Python · SQL · Java · C/C++ · Bash
Modeling & Evaluation	PyTorch · TensorFlow · A/B Testing · Uncertainty Estimation
LLM & GenAI	RAG · LangChain · LLM Guardrails · Embeddings · CLIP · Vector DBs (Pinecone)
Data & ML Pipelines	PySpark · Databricks · MLflow · Metaflow · Weights & Biases
Backend & Deployment	FastAPI · Django · Docker · Kubernetes · Terraform
Cloud & CI/CD	AWS (S3, EC2, SageMaker) · GitLab CI

Education

Master of Science in Computer Science, State University of New York, Buffalo, NY 2014

B. Tech. in Computer Science and Engineering, IIIT, India 2010

Publications

 [Google Scholar profile](#)

[Inferring Latent Attributes of an Indian Twitter user using Celebrities and Class Influencers](#)  ACM Hypertext 2015

[Inferring gender of a Twitter user using celebrities it follows](#) CORR 2014

[Architecture for Automated Tagging and Clustering of Song Files According to Mood](#) IJCSI, 2010

Open Source & Community

 Organizer @ MUFin	Program committee and paper reviewer, Modeling Uncertainty in the Financial Sector (AAAI 2023, ECML-PKDD 2022)
 Lotion	Unofficial Notion.so Desktop app for Linux (2K+ GitHub stars / 60K+ Clones & Downloads)
 Romadeva	Roman-to-Devanagari transliteration (Used by Translators Without Borders)
 Quena	Question-answering over 1.6M Wikipedia documents; query parsing and popularity-based ranking. (Apache Solr, NER, POS tagger)